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Combining Mechanical, Electrical and Thermal Energy Conversion for Ecological Vehicle Energy Harvesting Concepts

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Abstract—In comparison with common combustion motor driven vehicles future electric driven vehicles vary more in their reachable range because of the high drive train efficiency and the different energy demand requirements in different drive conditions. One of the main issues to improve the overall range is the minimization of energy consumption. On the one hand an adapted energy management strategy is needed to achieve a more comfortable and acceptable equalization in the range. On the other hand it is important to recuperate mechanical, electrical and thermal energy, both inside and from the vehicles wherever possible. For example, it is thinkable to use the kinetic energy of braking vehicles outside the vehicle to power traffic lights. If a combustion process is used, a lot of waste thermal energy can be re-used from the tailpipe. Or solar energy may be used. If a vehicle has a varying duty-cycle it makes sense to adapt the powertrain to suit the power need. Whatever the available energy, it has to go hand in hand with minimizing the energy consumption. This paper is based on the knowledge and combination of future technologies [2], which are useful to collect and convert energy coming from different sources within a vehicle.

Keywords—energy harvesting, waste heat conversion, thermoelectric generator, thermal storage, thermochemical storage, phase change material, shock absorber, magnetocaloric refrigeration.

I. INTRODUCTION

The German Aerospace Center (DLR) with its Institute of Vehicle Concepts is working on the development of future road and rail vehicle concepts in its transport research program. The aim is to provide efficient, sustainable and affordable mobility for tomorrow. The societal challenges are the climate protection, an increasing urbanization, the use of new energy sources and an increasing automation and digitalization process. In the

DLR-internal project Next Generation Car (NGC) three vehicle concepts with different application profiles were designed to show the potential of future technologies [1, 2]. These conceptualized vehicles are mainly driven electrically and are called the Save Light Regional Vehicle (NGC-SLRV) with its focus on light weight structures, the Urban Modular Vehicle (NGC-UMV) characterized by a high grade of modularization, and the Inter Urban Vehicle (IUV) for luxury long range usage, see figure 1.



Fig. 1: DLR-Next Generation Cars: Save Light Regional Vehicle (left), Urban Modular Vehicle (middle), and Inter Urban Vehicle (right) [1]

The AE-Group (AE) is specialized in advanced electromagnetic solutions, consultancy and custom-made drives, such as electric motors for automotive, high-tech and medical applications including tubular/linear actuators. AE develops, in-house, the mechanical, electromagnetic, electrical, electronic and thermal aspects of the electric drives and can perform research and production on the integration of motors within the application of our partners. AE has in-house facilities to produce and test electric machines and generators, where several testing facilities are available and custom test-benches are continuously being designed and built to serve the needs of our customers. The specific fields of expertise are: Electromagnetic, hydraulic, electronics and thermal modelling of electric drives, development of electric drive optimization software and a lot of experience in state-of-the-art electric drive technologies and powertrain control strategies. These mechatronic actuators, like electric machines, are going hand in hand with a world-class medical grade certified quality management system to assist with innovative future technologies like direct-drive

generators either connected to a low speed, e.g. 200 rpm, propeller shaft or a high speed, e.g. 30,000 rpm, thermodynamic turbine shaft.

The still unused potentials of energy storage and energy conversion in electric vehicles differ in their different physical manifestations. There are mechanical, electrical, thermal and chemical kinds of energy that can be stored or suitably converted into a usable form and that must be electronically controlled by an effective energy management (figure 2). A lot of engineering disciplines are required. To investigate the potential of re-use of waste heat or of collecting the energy from different sources, the effects of different thinkable mechatronic technologies shall be described and compared in the following chapters. One main topic describes waste heat recovery concepts as thermoelectric generators, Clausius Rankine cycle converters and metallic latent heat storage systems. The next topic looks to alternative refrigeration concepts as hydrogen based thermal storages and magnetocaloric refrigeration concepts. The third topic is focused on concepts by integrating auxiliary units directly to the drive shaft as the Two-in-One motor concept and parallel hybrid drives and the fourth topic describes mechanical energy absorber concepts like vibration or shock absorbers. The different technologies are described by their physical basic behavior and the known reached effects coming from literature review and test results. A comparison of the reachable energy saving potential will be focused on an urban used car like the Urban Modular Vehicle whereas the New European Drive Cycle (NEDC) is one of the essential requirements.

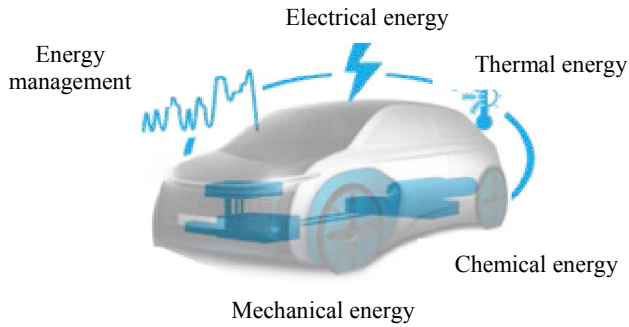


Fig. 2: Different kinds of energy in a vehicle

II. WASTE HEAT RECOVERY CONCEPTS

The high grade of vehicle modularization also allows thinking about the potential of range extender concepts. If an Internal Combustion Engine (ICE) is still used for this, heat on a relatively high temperature level exists and may be recovered.

A. Thermoelectric generator [2, 3]

Thermoelectric energy conversion was used for power supply systems of satellites in interstellar operation [2]. In

all cases where waste heat is produced at a high temperature level, for example from combustion processes such as electric vehicles with range extender, the use of thermoelectric energy conversion is possible.

The basis is to use the Seebeck effect, which states that a thermoelectric material generates an electric voltage if it is subjected to a temperature difference.

According to its internal electrical resistance, the voltage source then may provide electrical power. The voltage U , produced by the temperature difference ΔT , is

$$U = S \cdot \Delta T, \quad (1)$$

where S is the Seebeck-coefficient of the specific material. Thermoelectric materials are characterized with the ZT -value as a dimensionless number which includes the quality of the Seebeck-coefficient S and the application temperature T .

$$ZT = S^2 \cdot \frac{\sigma}{\kappa} \cdot T \quad (2)$$

It is important to recognize the opposite requirement in itself to the material, to have a good electrical conductivity σ , at the same time with a poor thermal conductivity κ . These characteristics have semiconductor materials which are optimized by means of doping processes. Their mechanical behaviour is similar to that of ceramic. The efficiency results in an equation which depends on a high Carnot efficiency $((T_h - T_c)/T_h)$ and on a high ZT -value:

$$H = \frac{P_{el}}{Q} = \frac{T_h - T_c}{T_h} \cdot \frac{1}{\frac{1}{4ZT} + 2 - \frac{1}{2} \cdot \frac{T_h - T_c}{T_h}} \quad (3)$$

P_{el} is the electric power, Q the thermal Power, T_h the temperature at the high side and T_c the temperature at the cold side.

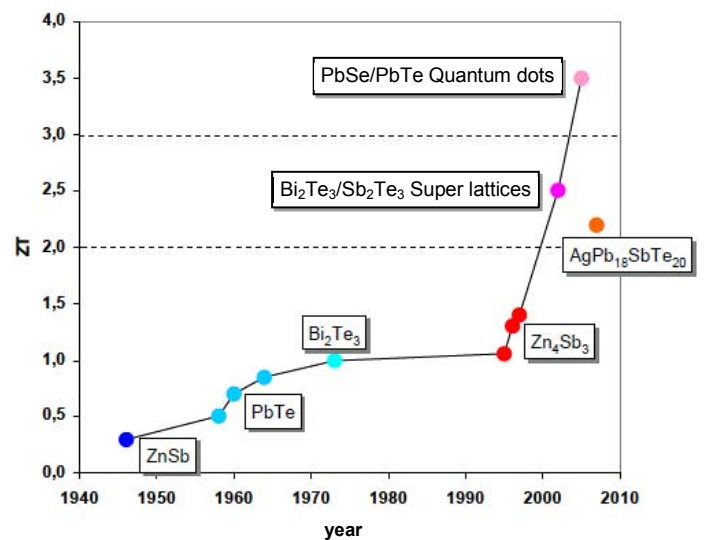


Fig. 3: Thermoelectric materials characterized by their ZT -value [2]

Figure 3 shows the ZT-values of typical thermoelectric materials depending on the investigation timeline. For the demonstrators in the automotive branch PbTe and Bi₂Te₃ are the most used materials. Figure 4 shows the design of two thermoelectric generators reaching an electric power up to 200 W at an operation point of driving a car with a gasoline motor at 130 km/h [2, 3].

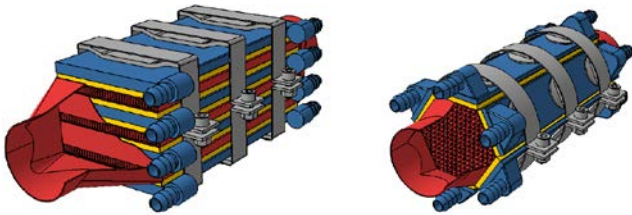


Fig. 4: Design of thermoelectric generators [2]

The thermoelectric material is shown in yellow color and is positioned between the exhaust gas heat exchangers (red) and the cooling water heat exchangers (blue). The thermoelectric generator shown on the left side in the figure divides the mass flow of the exhaust gas into several heat exchanger layers; in the design shown on the right only one heat-conducting layer is available. The design depends on a special power working point of the individual application. To protect the thermoelectric material against overheating, a bypass for the exhaust gas mass flow could be necessary. A maximum power point tracker converts the internal electric resistance to the load resistance. In addition, no moving parts are needed.

Although the energy conversion efficiency is only a few percent, this technology offers potential for heat recovery. Hybrid vehicles can benefit from the production of electrical energy from the exhaust gas. An energy saving potential of about 5% in the NEDC was estimated.

B. Clausius Rankine Cycle [4, 5]

A turbine can convert energy out of saturated steam generated by waste heat (if it contains sufficient caloric energy), see the Clausius Rankine Cycle in figure 5.

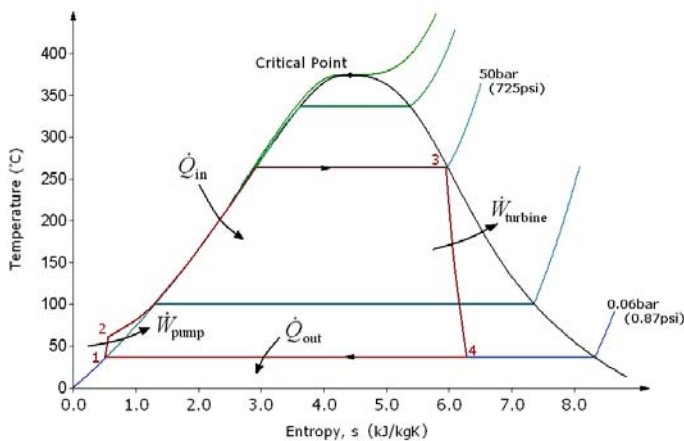


Fig. 5: Typical Clausius Rankine Cycle for 0.06 to 50 bar [4]

The resulting electrical power could be used for e.g. an electrical cooler and the waste heat to increase the cabin temperature. The basic idea, of such a turbine, takes the advantage of flue gasses, as shown by the Clausius Rankine Cycle in figure 5. It takes the heat that is converted by the turbines into electricity instead of cooling them down and wasting them through the tail pipe.

The so-called Green Turbine waste heat recovery (WHR) system uses a heat exchanger system that captures this heat and converts it into steam (200 °C and a pressure of around 10-12 bar abs.) in order to drive one or more 2.5 or 15 kWe Green (steam) Turbines. The steam turbine, an alternator and an inverter convert about 10-15% of the energy within the waste heat into useful electricity. The steam rotates the two turbines that have direct-drive high speed, ca. 30,000 rpm, generators and a reduced water temperature of 45 °C is a by-product of which some is being pump back into the system to make sure a minimum amount of heat is used to generate new steam.

In general, two types of steam turbines for Combined Heat and Power (CHP) and WHR can be distinguished, namely condensing turbines with steam extractions and backpressure turbines with or without extraction. The Green Turbine™ is designed to achieve the best energy transformation by utilizing two turbine stages in the condensing mode of operation, as shown in figure 6. As such, considering the operate modes of the turbine in the CHP mode the efficiency is high over a large variation of power levels.

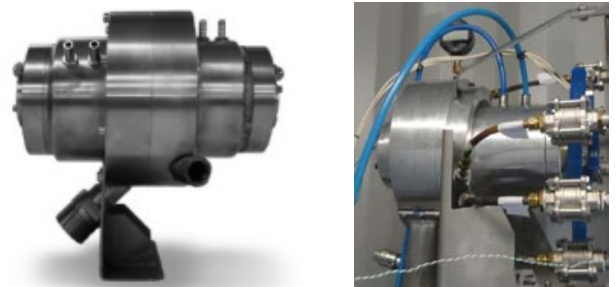


Fig. 6: Waste heat conversion, 2.5 & 15 kWe Green Turbines [5]

In this two turbine configuration, steam brings and keeps the first turbine in rotation. To this end super-atmospheric steam is supplied to one or more nozzles, which sprout the steam at supersonic speed in the channels of the first turbine. This rotating first turbine reverses the tangential movement of the steam and sprouts it into the second channel of the second turbine wheel, so as to utilize most of the energy present within the steam. Following, the steam will be condensed, i.e. it becomes water, which is then fed back by a pump to the, for example, heat exchanger to, once more, generate super-atmospheric pressure medium (steam) therefrom.

The typical Sankey diagram of figure 7 shows that 41% of the exhaust gas of a diesel engine is currently not

utilized by the engine itself and therefore thrown away. This could of course also be a range extender or any other source of heat, e.g. generated by the sun.

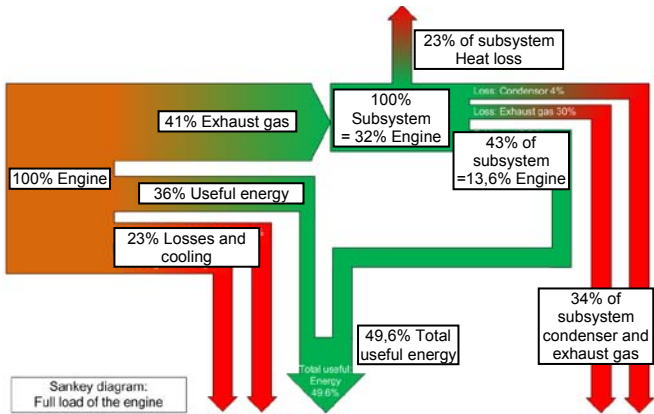


Fig. 7: Typical Sankey diagram showing that 41% of the exhaust gas of a diesel engine is wasted [5]

C. Metallic Latent Heat Storage systems

Future road vehicles will have partially or fully electrified drive concepts. In comparison with ICE powered vehicles, there is comparatively little waste heat available for air conditioning and temperature control purposes. This lack of waste heat, with simultaneously increasing demands on efficiency and comfort, places great challenges on future thermal management systems. This is particularly evident in the thermal management of fully electrified vehicles on cold winter days. Standard PTC heaters use electric energy from the traction battery to convert it into heat. In the process the drive system loses this energy which results in a range reduction of the vehicle by up to 50%.

To overcome this deficit, heat storage systems using metallic Phase Change Materials (mPCM) can be integrated into electric vehicles. mPCM are used to store thermal energy at a high temperature level of up to 600 °C.

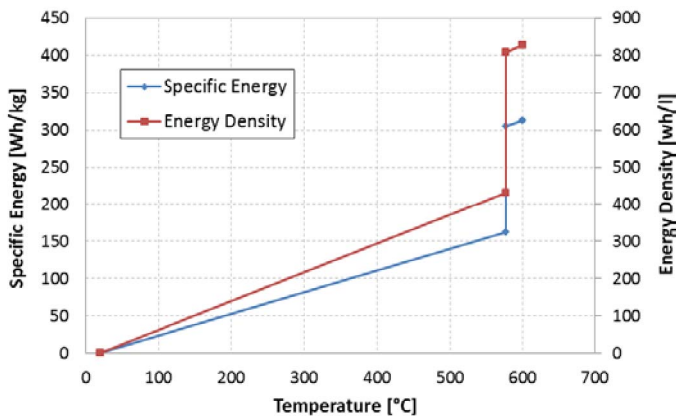


Fig. 8: Specific Energy and energy density in the temperature range from 20 °C to 600 °C of the metallic Phase Change Material (mPCM) AlSi12

With the use of mPCM, both the sensible and latent thermal energy is used to enable a high specific energy and energy density (compare figure 8). As phase change, the conversion between solid and liquid is used.

As quite generally known, conventional phase change materials like salts or paraffins suffer from their low thermal conductivities which are generally less than 1 W/mK, limiting thermal energy storage applications regarding high thermal heat transfer rates. In contrast, mPCM show thermal conductivities of a magnitude of about 100 times higher with values in the range from about 35 W/mK up to 200 W/mK. As a disadvantage of mPCM, relatively high heat losses as well as the limited thermal compatibility to components of the drive train such as the battery (temperature level of about 0 °C to 40 °C) due to the high working temperatures of mPCM up to 600 °C have to be mentioned.

By using both mPCM and conventional PCM in one thermal energy storage, DLR has created a storage concept called DuoTherm that combines the benefits of both storage materials. The basic principle of the combined heat storage system DuoTherm is the energy storage at two different temperature levels. This is realized by the concentric design of the storage unit (comparable to structure of the earth). In the center, a high amount of thermal energy is stored at high temperature level in mPCM. Enclosing the hot core, the outer shell of the system is storing thermal energy on a lower temperature level in conventional PCM. The basic concept to realize this storage design is shown in figure 9.

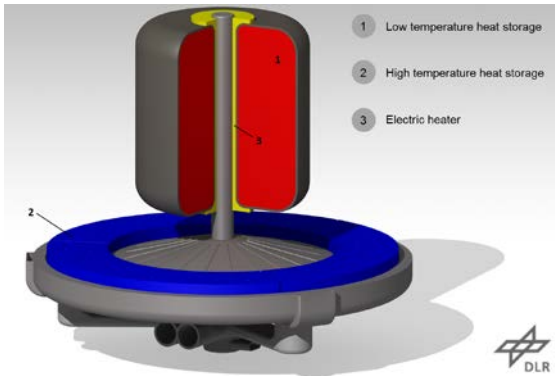


Fig. 9: Concept of the combined thermal storage system DuoTherm (mPCM red, PCM blue, heater yellow)

Based on this dual functional design, future thermal energy storage systems will have great advantages over singular storage systems. In the course of this, thermal losses of the hot core can be reduced or more specifically be utilized by the low temperature storage unit. In addition, combining two heat storages enables the needs-based maximization of the specific storage density while offering ideal integration capabilities into the overall system.

Overall efficiencies of electric vehicles can be risen by storing heat losses of the charging unit or the battery in the low temperature storage section. Range can be increased by pre-conditioning of the high temperature storage section. With the use of the stored energy for heating the vehicles cabin, electric energy stored in the battery can be saved and the overall range of the vehicle extended. With a system weight of about 10 kg it is possible to store a thermal energy of 1.25 kWh, partly pre-conditioned and regulated by an external electric power electronics. From today point of view this is similar to battery storage and leads to an increased range of 50% if an urban vehicle is used half an hour with an average drive power of 5 kW in urban use.

III. ALTERNATIVE REFRIGERATION CONCEPTS

A. Hydrogen-based thermal management [6]

Another research focus in the area of thermal energy converter is to investigate the usability of thermochemical reactions on the basis of hydrogen [6]; this can also belong to the topic waste heat recovery. For example, metal hydrides (MeH) can be used for the vehicle air-conditioning when the exothermal or endothermal character of the reaction is used. In this approach, on the one hand, the characteristic of the endothermic desorption of hydrogen in MeH is applied to generate a cooling effect and on the other hand, the bonding of hydrogen in MeH with the exothermic reaction is used for heating. On material basis, MeH enable both very high reaction rates and reaction enthalpies. Thus, only low MeH masses are required and small systems are aspired. According to figure 10, a hydrogen-based thermal energy converter uses two identical and compact metal hydride reactors for continuous cooling and heating in fuel cell vehicles. The two reactors are integrated between the hydrogen pressure tank and the fuel cell. In two half-cycles, these reactors are alternately charged and discharged with hydrogen.

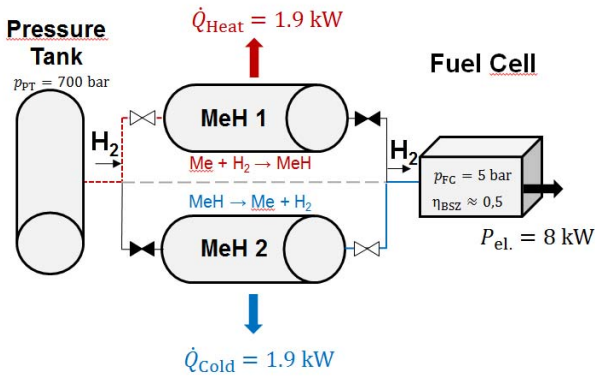


Fig. 10: Theoretical basis of a cooling and heating system based on metal hydrides [6]

In the so-called regeneration half-cycle, a metal hydride reactor (in this example MeH 1) is charged with

hydrogen from the hydrogen pressure tank and heat is released [6]. In the following cooling half-cycle, the hydrogen is subsequently released to the fuel cell by absorbing heat at a low temperature level and thus cold is generated. A second metal hydride reactor (in this example MeH 2) alternately passes the same two half-cycles for continuous cooling and heating.

The driving force of the system is the on-board existing pressure difference between the pressure tank ($p \approx 350\text{--}700\text{ bar}$) and the fuel cell ($p \approx 5\text{ bar}$). In state of the art vehicles, the hydrogen pressure is throttled down to the fuel cell level and the compression work is so far wasted.

Figure 11 shows the schematic integration concept of the hydrogen-based thermal energy converter into an electric vehicle which is equipped with a retrofittable modular fuel cell range extender. In this figure four thermal circuits are shown. The green one is the conventional air conditioning refrigerant circuit which is not touched but which can be calculated in a modified kind in the simulation model. The reactors are integrated into the low-temperature coolant circuit (battery, power electronics an electrical machine, drawn in blue) and the cabin cooling circuit (in black) by using three-way valves. Therefore, the heat and cooling generated by the reactors can be used to match the temperature of the vehicle's battery as well as the air-conditioning of the cabin. The red drawn coolant circuit provides the fuel cell with heat, using a PTC, or cools it down using a heat exchanger in the front end of the vehicle. In this example it is not coupled to the MeH reactors.

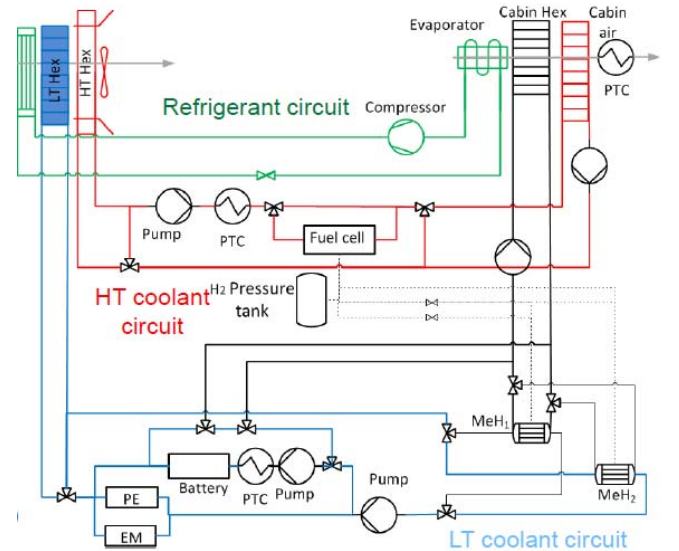


Fig. 11: Concept for integration of thermochemical storages into the cooling circuits of an electric vehicle with a fuel cell range extender to produce heat, coldness and electricity [6]

In a heating mode, the waste heat of the reactors is provided to warm the cabin and the battery at cold temperatures. In a cooling mode, the cold generated by the

MeH reactors is used to cool both the cabin and the battery pack.

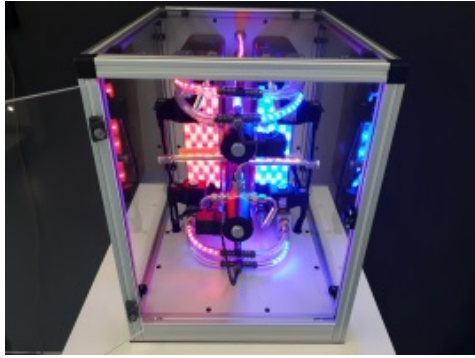


Fig. 12: H₂-based two storage energy converter without fuel cell

To offer a solution for battery electric vehicles as well as fuel cell vehicles, a modular concept for an electrical and thermal energy converter has been designed at the DLR. The so-called air-conditioning and auxiliary power unit (A/C-APU) is the combination of the metal hydride reactors (figure 12) with the fuel cell and the hydrogen pressure tank integrated in the electric and thermal circuits of a vehicle. This unit can be used as an auxiliary power unit (or a range extender) with the ability to provide heat, cold and electricity for battery electric vehicles or fuel cell driven electric vehicle (FCEV). Using the additional energy carrier hydrogen, with the A/C-APU increase of the vehicle range can be achieved. Assuming that the energy consumption of an urban vehicle in the New European Driving Cycle (NEDC) is equal to 14 kWh/100 km, an energy saving potential of about 0.308 kWh (2.8 kWh/100 km) in comparison to a state-of-the-art fuel cell vehicle is expected, since the cooling/heating power of the MeH reactors corresponds to 20% of the fuel cell electrical power [6].

In the case of an HVAC energy consumption of about 14.9 kWh/100 km at -10 °C and 7.8 kWh/100 km at 30 °C it is possible to achieve a reduction of the HVAC energy consumption of about 19% und 18%, respectively, in comparison to a state-of-the-art fuel cell vehicle.

B. Magnetocaloric refrigeration [7, 8, 9]

On the way to the efficient and affordable air-conditioning systems for the mobile applications avoiding use of ozone depleting, greenhouse gases or hazardous chemicals as refrigerants the magnetic refrigeration technology is a great deal of interest [7].

Magnetic refrigeration utilizes the magnetocaloric effect (MCE), the temperature increase in certain materials when exposed to a magnetic field and reversible cooling down upon removal of the magnetic field. An expected coefficient of performance (COP) is significantly higher than one of conventional air conditioning systems. Figure 13 shows that alloys consisting of iron, silicon and

lanthanum with hydrogen produce a temperature level of near to 7 K in the interesting operating temperature range between 240 and 320 K (-30 until +50 °C) if they are disposed in a magnetic field with a flux density of 2 T [8].

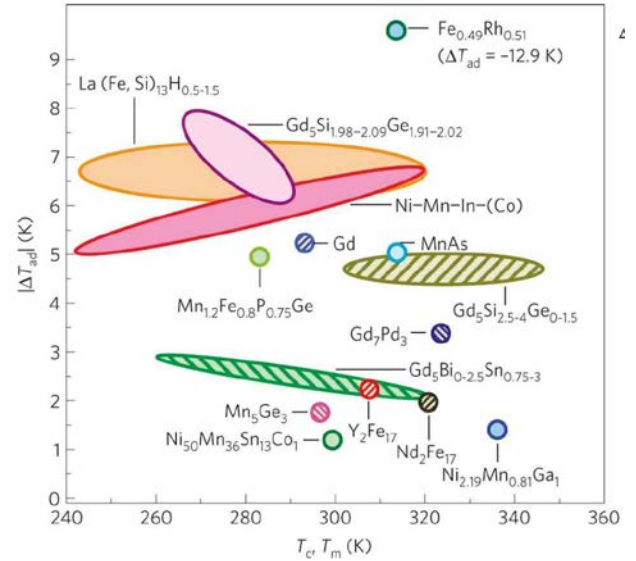


Fig. 13: Magnetocaloric material behaviour where the magnetic flux density is 2 T [8]

Under real conditions a flux density of 1.5 T is more realistic, the kind of the material blending concentration leads to a large bandwidth of materials with corresponding operating temperatures as illustrated in figure 14 with lines of different color [9].

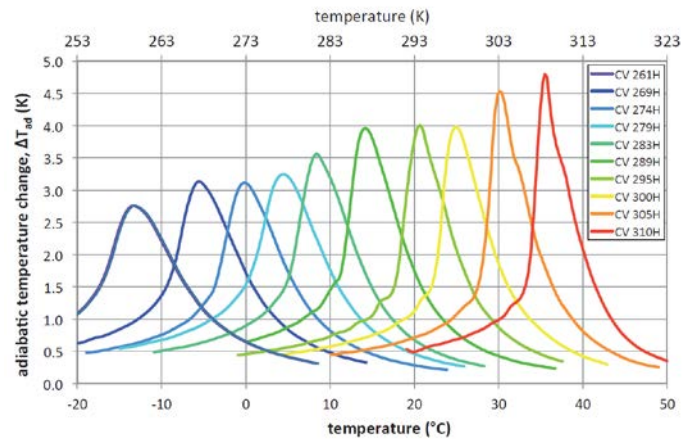


Fig. 14: Lanthanum-iron-silicon-hydrogen magnetocaloric material behaviour where the magnetic flux density is 1.5 T [9]

Cascading several alloys allows a summation of the single temperature steps so that it is possible to create a temperature difference of 40 K what is enough to use it as a refrigeration technology. One of the challenges is to transfer the heat or cold to an external heat exchanger by changing the coolant circuit between the two temperature levels. The other challenge is to produce a relatively high magnetic flux density with a minimum of material effort.

In figure 15 a demonstrator design shows the coolant channels in the green coloured end plate of the cylindrical machine in which the magnetocaloric material and the coolant liquid is in contact. The electromagnetic circuit consists of a rotating permanent magnet excited inner part and a magnetic yoke made of iron in the outer part of the cylinder.

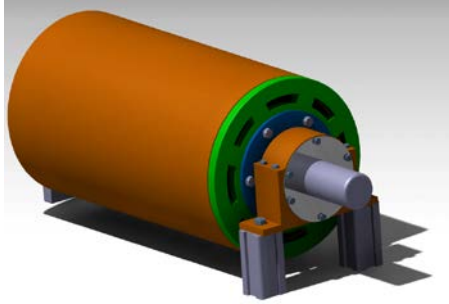


Fig. 15: Demonstrator design of a magnetocaloric energy converter [7]

Expecting a COP higher than one of conventional air conditioning systems there might be an additional potential to save energy in the total vehicle system. Regarding the high effort of using rare earth material and a relatively slow turning speed because of the thermal delay time the third challenge is to search for a further combination of electromagnetic circuit function, thermal management system and the integration of the electronic control unit in one producible unit.

In the NEDC the auxiliary units usually are switched off. In a real drive cycle where the air conditioning unit is switched on, an energy saving potential of 5-10% in comparison to a conventional air conditioning unit may be estimated.

IV. DIRECT COUPLING OF AUXILIARY UNITS TO THE DRIVE SHAFT

A modular drive train concept as it is part of a modular vehicle like the NGC-UMV may consist of one central motor or a combination of two different sized motors. As it is known from Hybrid Electric Vehicle drive trains the combination of two motors with different torque-speed characteristics can decrease the total energy consumption.

A. Two-in-One Motor Concept [10, 11]

Auxiliary drive units such as blowers or pumps in purely electric vehicles today are supplied by electrical energy that goes through an extensive energy conversion chain. Charger, inverter, the charging-discharging behavior of batteries, DC/DC converters even electromechanical converters and gears are always characterized with their maximum efficiency point. Reducing the number of energy conversion stages is an essential step towards more efficient use of high-quality electric power [10]. If the air conditioning compressor

could be directly coupled to the rotating wheels during the braking events, the overall efficiency must rise.

The other aim is to use the two electric machines to drive the vehicle always in their best efficiency ranges [11]. One operation strategy is to switch over the two motors 1 and 2 according to the condition of the highest efficiency for the operating point, see figure 16 as an example for the first 200 s of the New European Drive Cycle (NEDC) using two permanent magnet excited synchronous machines.

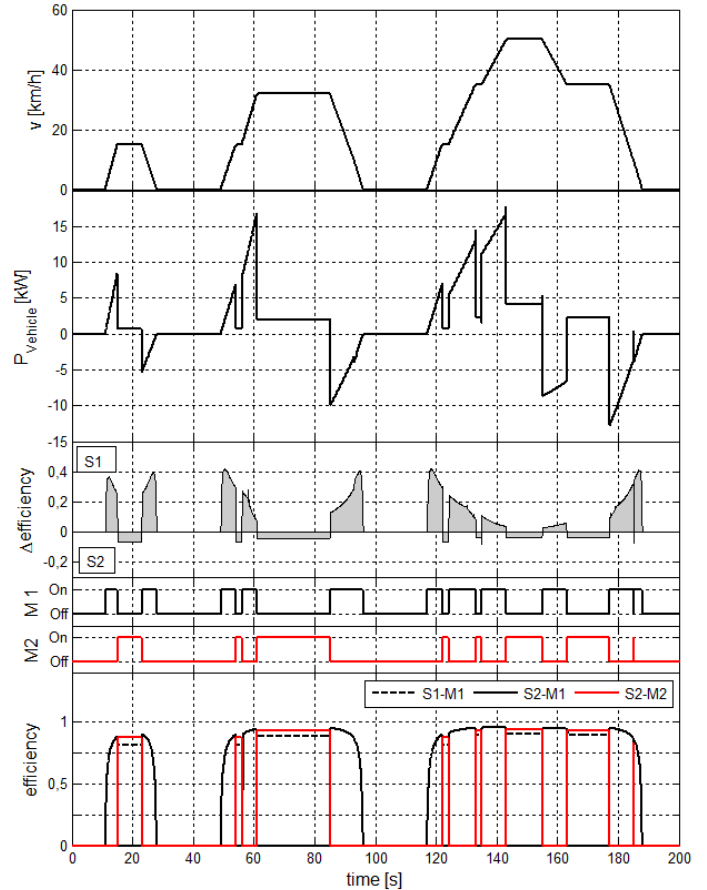


Fig. 16: Operation strategy by using the motor with the highest efficiency for the first 200 s of NEDC (for better view) [11]

The motor 1 with a maximum power of 50 kW delivers the high torque for acceleration while the smaller motor 2 with a maximum power of 10 kW is used to drive at constant speed. The switching condition between M1 and M2 belongs to the efficiency difference of each motor. Figure 17 shows the strong dependency of the energy consumption on the selected driving cycle, and because of the high efficiency in the case of electric drives this has a much greater effect than in the case of using combustion engines and which also contributes to the known uncertainty with respect to the range expectation. In figure 17 four drive cycles and four different combinations of permanent magnet excited synchronous machines

(PSM) and asynchronous machines (ASM) were used to calculate the energy saving potential in comparison to the usual case where only one traction motor is available. In addition to the NEDC and the WLTC two real drive cycles were used, where ST-LA (Stuttgart-Lampoldshausen) represents a very low energy consumption on a national road and a mountain descent way, whereas LA-ST a high power demand on highway including a mountain climbing way represents. The difference in the altitude is 200 m.

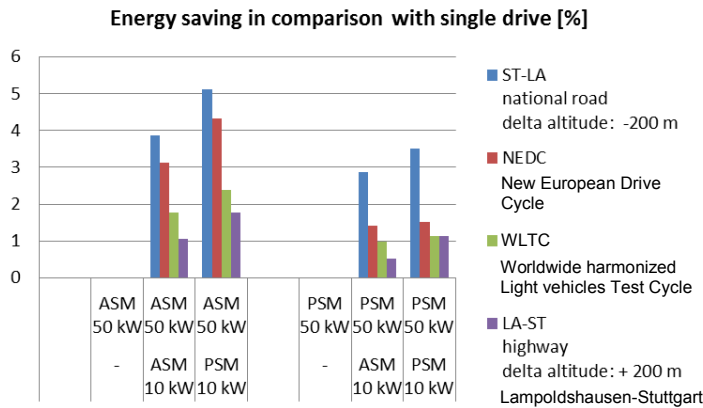


Fig. 17: Energy saving potential at different drive cycles [11]

If the vehicle is always used on the highway and requires high power, it may be discussable if a two motor concept will be helpful. But if the vehicle shall satisfy all consumer requirements, urban and interurban use, the combination of two different motors has the potential to increase the total efficiency range. As an average value an energy saving potential in the NEDC between 1% and 4% can be estimated from figure 17.

Figure 18 shows the design of a Two in One Motor demonstrator [11]. The larger drive motor M1 is directly connected to the driven shaft. The smaller motor M2, called compressor motor, is connected to the air conditioning compressor and may be coupled to the drive motor with an electromagnetic clutch. A planetary gear reduces the speed of the compressor motor.

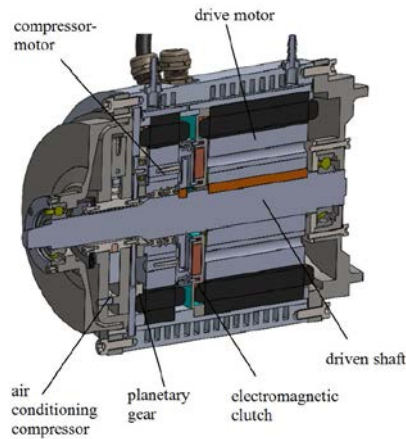


Fig. 18: Two in one motor [10]

B. Hybrid electric drive [12, 13, 14]

Looking forward to any vehicle which must not be a passenger car, ICE-electric hybrid drives still have a high energy saving potential. To maximize the return on investment of any energy harvesting system, the operational profile of either road vehicle or ship will always remain of utmost important. This is true for road vehicles, however also for ships, where the ship load can be powered from a generator that is placed around the shaft leading to the propeller.

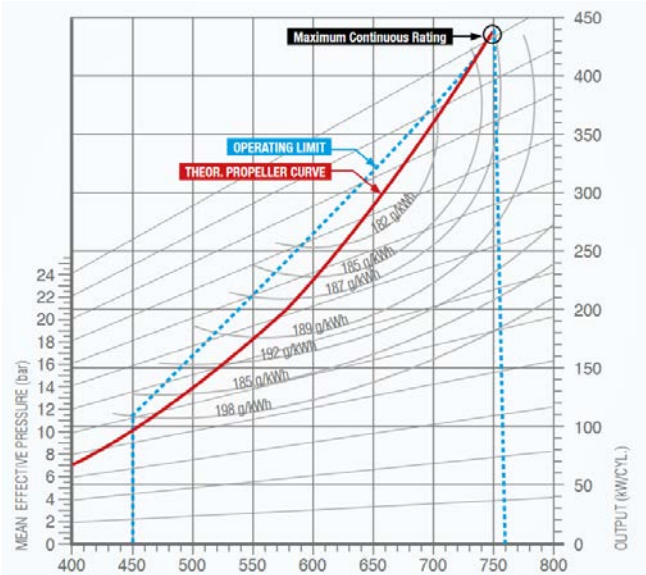


Fig. 19: Optimising engine efficiency [12]

To detail this in more details, ships are might steam at nearly full speed from point-to-point, however extended periods of time are spend periods at slower speeds. For example, offshore vessels spend several hours a day on dynamic positioning or standby operation, or anchorhandlers where full bollard pull is only used intermittently can benefit from a direct-drive shaft generator to power the auxiliary systems [13].

Also future inland ships require power systems with improved fuel economy, hence reduced emission, while they are able to meet the power demand. With the use of power electronic converters, shipboard distribution systems can be supplied from a direct-drive shaft connected electric machine.

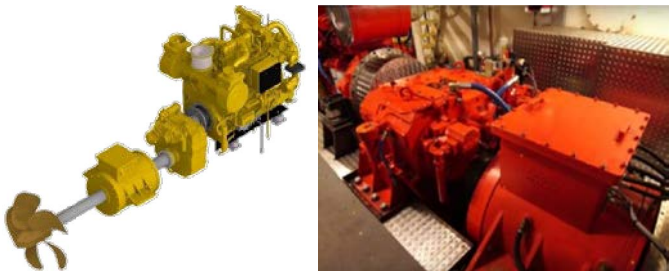


Fig. 20: Direct-drive electrical generator to a propeller shaft [12]

This electrical machine can also be used to power the ship during low power demand situations, like canals, harbors, etc. in combination with a battery. A clutch is present to decouple the electrical machine from the “main” diesel engine.

In this manner, rather than starting a separate diesel-generator set, the main diesel is loaded a little extra to allow for the hotel load on the ship, which can be rather high in case of containers that require cooling and/or cargo that require heat. For example, figure 19 shows the fuel consumption based on engine rpm per cylinder and figure 20 shows the drive shaft of the total system. For example, based on a 200 kW output, fuel consumption can be reduced from 198 g/kWh to 188 g/kWh. This translates to a 5 per cent fuel saving. Of course, does the resulting fuel efficiency depend on many variables, e.g. ship type (length, width, hull profile, etc.), trajectory (river, canals, depth, etc.) and load-cycle (frequency and magnitude of accelerations, duration of the stops, distances to be covered, etc.). However, on average this technology increases energy efficiency. For some specific journeys of inland ships, a fuel saving of even 25% has been reported [14].

V. MECHANICAL ENERGY ABSORBER CONCEPTS

A. Vibration energy absorber [15]

Most vibration based energy harvesting systems are used to power ubiquitously deployed sensor networks and mobile electronics. These systems scavenge power from their surrounding usually in the range of milliWatts. However, also systems that can harvest 10’s or even 100’s of Watts can be utilized that are vehicle powered. A proposed system is shown in figure 21 [15].



Fig. 21: Mechanical movement energy converter system [14]

As sustainable living gets an increasing foothold in people’s lives, unobtrusively tapping some fraction of energy available from every day activity will gain in popularity. In the last decades, energy harvesting has grown from long established concepts into devices that can be used to generate electrical energy from a variety of sources including thermal, solar, strain, inertia, etc.

The vehicle experience application considered requires a power level of a 100’s of Watts to power a LED lighting system and open a parking garage gate. Further, solar cells can be incorporated in the top layer if available. A simplified harvesting model [15] consists of a tile suspended by springs with mechanical springs that return the tile in the top position. The electrical generator acts as a damper. In this system, the linear vertical motion of the moving part of the tile is converted to rotary motion by a large gear ratio where the energy is harvested with a high efficient brushed DC generator, e.g. around 50% has been achieved with this system [15].

VI. RESULTS

TABLE 1: SUMMARY OF THE ENERGY SAVING POTENTIAL OF EACH TECHNOLOGY

<i>Technology</i>	<i>Qualitative estimated energy saving potential</i>	<i>Remark</i>
Waste heat recovery system		
Thermoelectric generator [2, 3]	Getting energy from the exhaust gas a range increasing potential of 1-5% is estimated	For ICE powered vehicles (range extender), no moving parts necessary
Clausius Rankine cycle [4, 5]	Getting energy from the exhaust gas a range increasing potential of 10% is estimated	For ICE powered vehicles (range extender)
Metallic latent heat storage	In an urban drive cycle with a typical driving time of 30 min a range increasing of 50% is estimated	Preconditioning of the cabin or the drive train components possible
Refrigeration system		
Hydrogen-based thermal management [6]	The range depends on the size of the hydrogen storage and may reach more than 100%.	Preferred in fuel cell vehicles
Magnetocaloric refrigeration [7, 8, 9]	In comparison with an common refrigerant circuit a range increasing potential of 5-10% is estimated	High rare earth material effort
Direct driven auxiliary units		
Two in one motor [10, 11]	A range increasing between 1 and 5% is estimated related to the combination of themotor type and drive cycle	More effective under real drive conditions
Hybrid electric drive [12, 13, 14]	A energy saving potential up to 25% is possible depending on the grade of hybridisation	For ICE powered vehicles (range extender), several operating strategies possible
Mechanical energy absorber		
Vibration shock absorber [15]	Not applicable in a vehicle drive cycle	The energy might be collected outside the vehicle

Energy harvesting, and the use of energy, will be evermore important in the modern day society. In this respect, numerous technologies could be utilized as mentioned in this article. Here an overview is given, of systems that are employed in practice and conceptual ideas. We hope that this article will inspire other engineers to continue, increase and/or start to consider energy harvesting and minimization of the energy wastage, since every wasted Watt should be considered in order to approach a sustainable society. Table 1 shows the potential of energy saving or power delivery of each technology. It is clear, that this are estimated values which shall demonstrate that the end of vehicle energy saving investigation potential is not reached at the state of art.

Looking to the different technologies to convert or store energy there is a wide area of range increasing potential depending if there is a re-use of waste heat (5-10%) or if the cabin is pre-heated in combination with an additional thermal storage (50%) until using an additional energy carrier like hydrogen (more than 100%).

VII. CONCLUSION

Several energy harvesting concepts like waste heat recovery or refrigeration systems, dual storage and dual motor concepts even as mechanical energy absorber concepts are described in this paper. It is visible that the end of the research process in reducing the total vehicle energy consumption is not reached until now. From the scientific point of view there still exist a potential to save energy, also to store energy in thermal or thermochemical storages. Material research, the combination of the different engineering disciplines from mechanical to electronical issues and the intermodal energy and vehicle system operation strategy are in the focus of future research. Coming from the idea to demonstrate different energy harvesting technologies as they can be used in an Urban Modular Vehicle concept, the presented technologies are not limited only to road vehicles, they may also be used in other ecological vehicles as trains or ships.

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REFERENCES

- [1] J. Weimer, S. Schmid, M. Schier, F. Rinderknecht, G. Kopp, F. Köster, T. Bunte: *Next Generation Car – Technologies for future EVs*, Electric Vehicle Symposium EVS29, Montreal, Canada, 2016.
- [2] Schier, M.; Rinderknecht, F.: *Innovation Examples for Ecological Vehicles based on Aerospace Research*, International

- Conference on Ecological Vehicles and Renewable Energies (EVER 13), 27.-30. Mar. 2013, Monte Carlo, Monaco.
- [3] C. Häfele, M. Schier, S. Hahn, T. Weiler, and H. Friedrich: *Experimental analysis of a test vehicle in terms of exergetic potentials and retroactive effects on the vehicle system at the integration of thermoelectric generators*, Thermoelectric Goes Automotive, pages 42-70. Expertverlag, Renningen. Thermoelectric Goes Automotive, 09.-10.12.2010, Berlin, Germany. ISBN 978-3-8169-3064-8
- [4] Sonntag, R.E., Van Wyllen, V.W., and Borgnakke, C., *Fundamentals of Thermodynamics*, John Wiley & Sons, New York, 2002
- [5] Paulides, J.J.H.; Post, E.; Post, J.; Encica, L.; Lomonova, E.A.: *Green turbine: A high speed double turbine solution for sustainable energy harvesting from waste heat*, 2015 Tenth International Conference on Ecological Vehicles and Renewable Energies (EVER 15), 1-7, 2015, Monte Carlo, Monaco
- [6] Nasri, M.; Bürger, I.; Schier M.; Friedrich, H. E: *Waste Heat Recovery for Fuel Cell Electric Vehicle with Thermochemical Energy Storage*, International Conference on Ecological Vehicles and Renewable Energies (EVER 16), 06.-08.04.16, Monte Carlo, Monaco
- [7] Lorenz, L. Kevlishvili, N.: *Designing of Halbach cylinder based magnetic assembly for a rotating magnetic refrigerator. Part 1: Designing procedure*. International Journal of Refrigeration, 73, Seiten 246-256. ELSEVIER. ISSN 0140-7007, 2016
- [8] Liu, J.; Gottschall, T.; Skokov, K. P.; Moore, J. D.; Gutfleisch, O.: *Giant magnetocaloric effect driven by structural transitions*, Nature Materials 11, 620-626 (2012)
- [9] Vakuumschmelze GmbH & Co. KG (VAC): *Magnetocaloric materials*, Hanau, Germany (2015)
- [10] Schier, M.; Kumar, S.: *Design of a 2 in 1 Motor to increase the Efficiency of Electric Vehicles*, International Conference on Ecological Vehicles and Renewable Energies EVER15, Monte Carlo, 29.03.-02.04.15, Monaco, 2015
- [11] Hubner, M.; Schier, M.; Kevlishvili, N.; Dietz, A.; Hörlin, S.: *Methodical Approach for designing electric propulsion systems containing two motors*, International Conference on Ecological Vehicles and Renewable Energies (EVER 17), 11.-13.04.17, Monte Carlo, Monaco
- [12] *Hybrid shaft generator propulsion system upgrade*, Rolls Royce brochure
- [13] Paulides, J.J.H., Djukic, N., de Roon, J.A. & Encica, L.: *Hybrid propulsion testing using direct-drive electrical machines for super yacht and inland shipping*, International Journal of Transportation Engineering and Technology, 2(4), 42-48. 2016.
- [14] C.D. Christophel (2011) “*Reduzierung der CO2-Emissionen durch diesel-elektrische Antriebe am Beispiel eines bestehenden Motorgüterschiffes*”, Workshop CO2-Emissionen der Binnenschifffahrt, pp. 1-11.
- [15] Paulides, J.J.H.; Janssen, J.W.; Encica, L.; Lomonova, E.A.; Smit, M.: *Human-Powered Small-Scale Generation System for a Sustainable Dance Club*, IEEE Industry Applications Magazine, Sep./Oct. 2011